

LAB 3 — WRITTEN JUSTIFICATION

Component Modelling & Architectural Pattern Selection | Self-Service Coffee Kiosk System

Name: Kishan Sangram **SRN:** PES2UG23CS277 **Section:** L

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice: The system is divided into presentation, business, and data/hardware responsibilities.

Reason 1: The touch-screen UI, order processing, payment handling, receipt printing, and menu/pricing data have separate responsibilities. This makes the kiosk easier to understand, test, and maintain.

Reason 2: The kiosk is a focused traditional application with a small set of predictable interactions. Layered Architecture fits the scenario without the operational complexity and network overhead introduced by Microservices.

Security Advantage: Payment processing is isolated in the Payment Service and accessed through a defined Payment API, reducing direct coupling between the user interface and payment logic.

Performance Benefit: Local component communication avoids unnecessary distributed network hops. Menu and pricing information can be accessed directly through the data layer.

Components & Interfaces

Components: User Interface, Order Manager, Payment Service, Receipt Printer, Menu Database.

Interfaces: Order API, Payment API, Receipt Interface, Database Query, Credit Card Interface, Printer Hardware Interface.